# Full-budget PENCIL is often unnecessary.

**65%** GPU budget

**matches or beats full PENCIL**
in 3 of 4 test regimes — with ~40% less peak VRAM on Cora HeaRT.

`s = 2×10 vs 2×20`

## THE QUESTION

**Can we spend PENCIL's GPU compute only where a cheap graph heuristic is uncertain?**

## OUR INFERENCE-ONLY METHOD

1. **Rank** — Adamic–Adar on CPU
2. **Shortlist** — top k candidate pairs
3. **Re-rank** — PENCIL with budget s

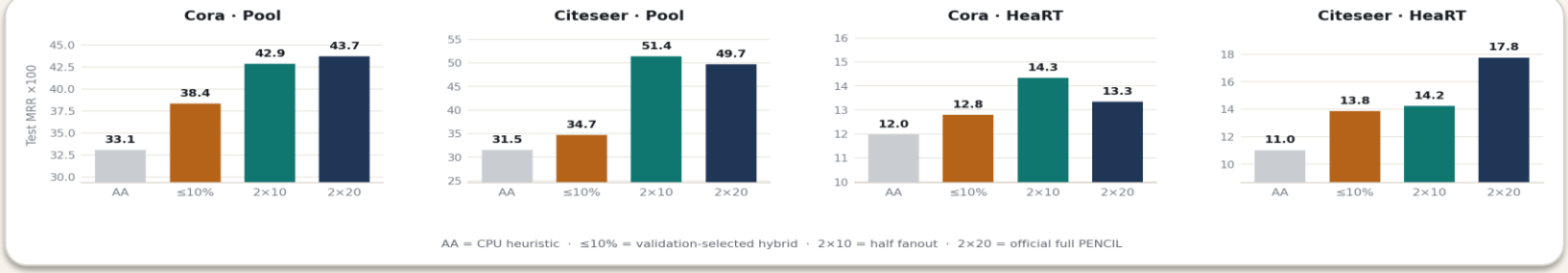
## MID-PROJECT STATUS

- ✓ **Reproduced PENCIL** Cora 44.00 · Citeseer 50.46
- ✓ **112 evaluation runs** 2 graphs × 2 protocols × budgets
- → **Next: final decision rule** latency/VRAM Pareto + final report

## HALF-FANOUT RESULT / TEST MRR ×100

| Cora · Pool | Citeseer · Pool | Cora · HeaRT | Citeseer · HeaRT |
|---|---|---|---|
| **42.9** | **51.4** | **14.3** | **14.2** |
| full: 43.7 | full: 49.7 | full: 13.3 | full: 17.8 |
| MATCHES | BEATS FULL | BEATS FULL | NEEDS FULL |

## THE FULL BUDGET IS NOT THE DEFAULT WINNER




**INTERIM LAW:** reduce neighborhood fanout first; reserve full context for the hard regime.

Amit Kumar · Ashmit Satpute · Arnav Kshirsagar · Pranav Shalya · Atharv Wankhede